

Surname

Name

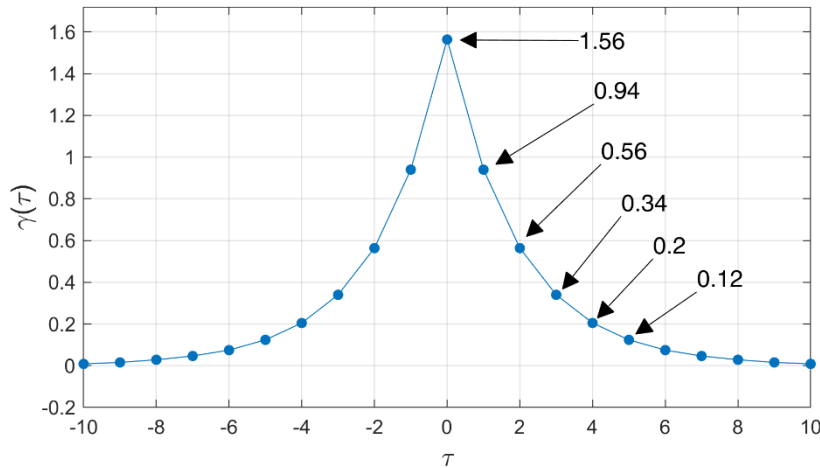
University ID Number

Signature

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- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 4. Additional papers will not be considered.
 - Clarity, order and precision are strongly considered for the final evaluation.
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1. [Identification]

Let $y(t)$ be a stationary stochastic process, whose covariance function $\gamma(\tau)$ is shown in the following Figure.



Consider the model class $\mathcal{M}_1: y(t) = \theta \cdot y(t - \eta) + e(t)$, where θ, η are model parameters and $e(t) \sim WN(0,1)$. Answer the following questions.

- Write the time-domain equations of the 1-step ahead predictor, as a function of the model parameters.
- When $\eta = 1$, identify the value of parameter θ .
- Repeat the identification of θ considering $\eta = 2$. Which of the two identified models is the best one? Motivate your answer.
- Consider the following model class $\mathcal{M}_2: y(t) = \theta \cdot e(t - \eta) + e(t)$: would it be a better model class choice, compared to $\mathcal{M}_1$? Motivate the answer.

Solutions

- a. First, the model class is written in the transfer function (filter) form:

$$y(t) = \frac{1}{1 - \theta \cdot z^{-\eta}} e(t) \rightarrow y(t) = W(z) e(t)$$

As the transfer function $W(z)$ is written in canonical form (provided that the value of the parameter θ is such that its poles are inside the unit circle), the 1-step ahead predictor is:

$$\hat{y}(t+1|t) = \frac{\theta \cdot z^{-\eta}}{1} y(t+1) = \theta \cdot z^{-\eta+1} \cdot y(t)$$

Its time domain expression results:

$$\hat{y}(t+1|t) = \theta \cdot y(t - \eta + 1)$$

- b. The identification of the parameter θ is done following the 1-step ahead prediction error minimization approach.

$$\bar{\theta} = \min J(\theta), \quad J(\theta) = E[(y(t+1) - \hat{y}(t+1|t))^2], \quad \hat{y}(t+1|t) = \theta \cdot y(t)$$

The cost function is expressed as a function of the covariance function $\gamma(\tau)$, yielding:

$$J(\theta) = E[(y(t+1) - \theta \cdot y(t))^2] = E[y(t+1)^2] + \theta^2 E[y(t)^2] - 2\theta E[y(t+1)y(t)] = \gamma(0) + \theta^2 \gamma(0) - 2\theta \gamma(1)$$

Setting its first derivative equal to zero allows to identify the model parameter

$$\frac{dJ}{d\theta} = 0 \rightarrow 2\theta \gamma(0) - 2\gamma(1) = 0 \rightarrow \bar{\theta} = \frac{\gamma(1)}{\gamma(0)} = 0.6$$

Notice that with the identified parameter, the model $W(z)$ is asymptotically stable, as required.

- c. Following the same prediction error minimization for $\eta = 2$ the value of the new parameter $\bar{\theta}$ is found.

$$\bar{\theta} = \min J(\theta), \quad J(\theta) = E[(y(t+1) - \hat{y}(t+1|t))^2], \quad \hat{y}(t+1|t) = \theta \cdot y(t-1)$$

$$\begin{aligned} J(\theta) &= E[(y(t+1) - \theta \cdot y(t-1))^2] = E[y(t+1)^2] + \theta^2 E[y(t-1)^2] - 2\theta E[y(t+1)y(t-1)] \\ &= \gamma(0) + \theta^2 \gamma(0) - 2\theta \gamma(2) \end{aligned}$$

$$\frac{dJ}{d\theta} = 0 \rightarrow 2\theta \gamma(0) - 2\gamma(2) = 0 \rightarrow \bar{\theta} = \frac{\gamma(2)}{\gamma(0)} = 0.36$$

Also in this case, with the identified parameter the resulting model is asymptotically stable.

To assess which is the most suitable model in describing the stochastic process, the value of the respective cost function for the identified parameter is compared.

$$\begin{aligned} J_{\eta=1}(\bar{\theta}_{\eta=1}) &= 1.56 + 0.6^2 \cdot 1.56 - 2 \cdot 0.6 \cdot 0.94 = 1 \\ J_{\eta=2}(\bar{\theta}_{\eta=2}) &= 1.56 + 0.36^2 \cdot 1.56 - 2 \cdot 0.36 \cdot 0.56 = 1.36 \end{aligned}$$

Since $J_{\eta=1}(\bar{\theta}_{\eta=1}) < J_{\eta=2}(\bar{\theta}_{\eta=2})$ the model having $\eta = 1$ is the best one. Additionally notice that for such model the prediction error variance (the value of $J_{\eta=1}(\bar{\theta}_{\eta=1})$) is equal to the process noise variance and no other model can perform better than this.

- d. The model class $\mathcal{M}_2$ should not be preferred over $\mathcal{M}_1$. There are two reasons that justify this conclusion:
1. As previously noted, since the prediction error of the identified model class $\mathcal{M}_1$ is equal to the noise variance, no model class can do better.
 2. The model class $\mathcal{M}_2$ represents a moving average stochastic process. For this class of stochastic processes, the covariance function is non-null only for lag values smaller than the process order. It can be easily seen from the given covariance function that this doesn't happen for the process under analysis, since its covariance function is always not null. Thus, the model class $\mathcal{M}_1$, which represents an auto-regressive stochastic process, is a more suitable choice.

2. [4sid/Stochastic processes]

Consider the dynamic system with input $u(t)$ and output $y(t)$, described by the following transfer function

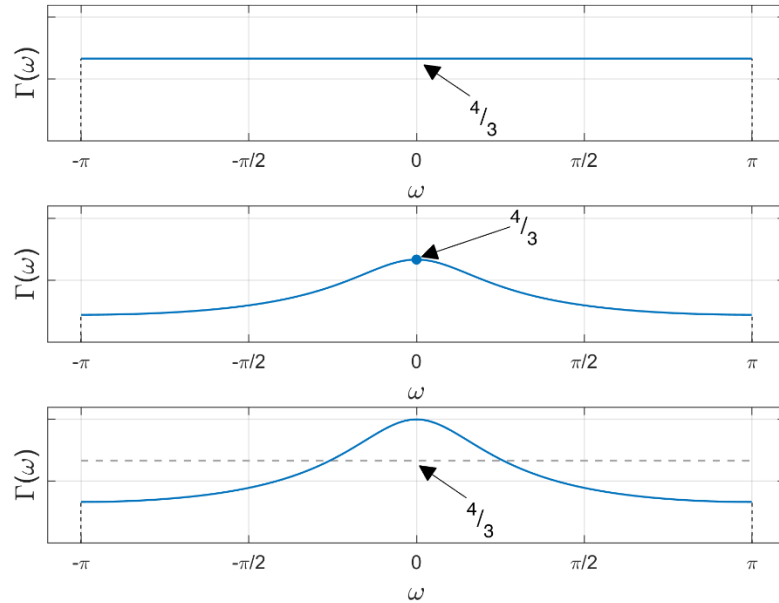
$$W(z) = \frac{z - \frac{3}{5}}{z^2 - \frac{11}{10}z + \frac{3}{10}}$$

and its impulse response samples $y(t)$:

$$y(0) = \dots, \quad y(1) = \dots, \quad y(2) = \dots, \quad y(3) = \dots, \quad y(4) = \frac{1}{8}$$

Answer the following questions:

- Compute the missing values of the system impulse response ($t = 0, 1, 2, 3$).
- Using the previous impulse response data, estimate the system order and compare it to the $W(z)$ one. Which considerations can be done?
- Using the impulse response data, write a suitable state-space representation of the dynamic system.
- Consider as system input a random noise $u(t) \sim WN(0,1)$. Knowing that the resulting stochastic process $y(t)$ has a variance of $Var[y(t)] = \frac{4}{3}$, which of the following representations of the power spectral density (PSD) of the process is the correct one?



Adequately motivate the answer.

Solutions

- By applying the long division between the numerator and the denominator of $W(z)$, the first samples of the impulse response can be computed:

$$\frac{C(z)}{A(z)} = z^{-1} + \frac{1}{2}z^{-2} + \frac{1}{4}z^{-3} + \frac{\frac{1}{8}z^{-2} - \frac{3}{40}z^{-3}}{z^2 - \frac{11}{10}z + \frac{3}{10}}$$

and the missing samples of the system impulse response are:

$$y(0) = 0, \quad y(1) = 1, \quad y(2) = \frac{1}{2}, \quad y(3) = \frac{1}{4}$$

- b. In order to estimate the system order using the impulse response data, the increasing order Hankel matrices are built:

$$H_1 = [\omega(1)] = 1 \rightarrow \text{full rank}$$

$$H_2 = \begin{bmatrix} \omega(1) & \omega(2) \\ \omega(2) & \omega(3) \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{bmatrix} \rightarrow \det = 0 \rightarrow \text{not full rank and not null}$$

So, the system order is $n = 1$. Given that the transfer function $W(z)$ has order $n = 2$, this means that it is written in a non-coprime form. In fact, it can be easily seen that there is a pole-zero simplification:

$$W(z) = \frac{z - \frac{3}{5}}{z^2 - \frac{11}{10}z + \frac{3}{10}} = \frac{z - 0.6}{(z - 0.6)(z - 0.5)} = \frac{1}{z - 0.5}$$

- c. Using the second order Hankel matrix, the state-space representation of the system can be computed:

$$H_2 = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{bmatrix} = O_2 \cdot R_2 = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} 1 & \frac{1}{2} \end{bmatrix}$$

The resulting state-space matrices are:

$$H = O_2(1, :) = [1]$$

$$G = R_2(:, 1) = [1]$$

$$F = O_2(1, :)^{-1} O_2(2, :) = \begin{bmatrix} \frac{1}{2} \end{bmatrix}$$

- d. The correct PSD is the last one. In fact:
1. The first diagram represents the PSD of a white noise. Considering that $W(z)$ is not an all-pass filter, the PSD of $y(t)$ cannot be equal to the one of a white noise.
 2. The variance of a stochastic process is related to its PSD according to:

$$\text{Var}[y(t)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma(\omega) d\omega$$

In other words, the variance is the average value of $\Gamma(\omega)$ over the interval $[-\pi, \pi]$. Since the second diagram has a maximum value of $\frac{4}{3}$ (in $\omega = 0$) it is not possible for the second PSD to have an average value of $\frac{4}{3}$. Thus, by exclusion, the third diagram represents the PSD of the process $y(t)$.

3. [4sid/Stochastic processes]

Explain the algorithm to estimate the state-space matrices starting from a noisy measurement of an impulse response (assume a strictly proper system)

Solutions

[See lecture notes](#)

4. [Kalman filter]

Describe (assumptions and results) the 1st and 2nd asymptotic theorems of Kalman Filter.

Solutions

[See lecture notes](#)